

POTHINENI SAI MADHU

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Educations

Vellore Institute of Technology (VIT), Chennai

B. Tech in Computer Science (AI & ML) - Focus on machine learning, data systems, and software engineering
Sri Chaitanya Jr College

Sep 2021 - Sep 2025

CGPA: 7.6

May 2019 - Jun 2021

Score: 85.2%

Apr 2018 - May 2019

CGPA: 7.5

Kendriya Vidyalaya Picket

Professional Experience

Innodatatics - (Data Scientist & Full Stack Intern)

Feb 2025 - May 2026

AI consulting firm building enterprise solutions in GenAI, data science, computer vision, and automation

- Designed and developed a multimodal intelligence monitoring platform for **TS Police** using NLP, Computer Vision, and Speech Processing, integrating three data modalities (text, image, and voice) to automate content analysis across monitored Telegram ecosystems.
- Built a real-time video ingestion pipeline using Kafka and RTSP to process **16** surveillance feeds continuously, supporting automated threat detection across 3 locations.

Robox Automation - (Web Development Intern)

Aug 2023 - Dec 2023

Product-driven startup building scalable web applications and infrastructure

- Optimized Django applications by improving database queries, caching strategies, and API response times, reducing page load latency by **30%**
- Implemented secure authentication and authorization mechanisms while collaborating with a 5-member team to deliver scalable, production-ready web applications

Lakshmi Collection - (Co-Founder / Operations Lead)

Jan 2018 - Present

Technology-enabled clothing retail business improving inventory control and day-to-day operations

- Engineered an automated retail management system for real-time inventory tracking, reducing stock tracking errors by **25%** and improving inventory visibility across daily operations.
- Executed smart storage chambers and introduced **24/7 surveillance**, improving inventory efficiency by **20%** and reducing loss

Research Project

Deepfake Detection using CNNs & ViT

- Developed **Vision Transformer (ViT)** deep-fake detection model, decreasing the rate of false positives by **12%** and reducing processing time per image by 0.3 seconds
- Achieved **94.6%** accuracy in detecting manipulated media, outperforming baseline CNN models by 35%

Smart Parking & Surveillance System

- Architected a Django real-time pipeline integrating Python, MySQL, and **multi-camera RTSP feeds**
- Integrated automatic **number plate recognition** (ANPR) for vehicle identification and event capture

AI Test-Case Generation Platform (Healthcare)

- Created a Vertex AI + LangChain framework for synthesizing and validating clinical test scenarios
- Applied RAG-driven context retrieval to improve accuracy and domain alignment, streamlining QA workflows and reducing manual review effort by **50%** while boosting overall data reliability

Interactive Deep Learning System for Car Damage Detection

- Improved YOLOv9-based deep learning system for real-time detection of **22 vehicle damage types** from multi-angle images
- Integrated React Insights to track model API latency and inference frequency, leading to refined monitoring and amplified production-grade reliability, reducing downtime by **15%** and improving response time by **50ms**

AI Assistant Platform (LLM + RAG + MCP)

- Architected an AI assistant platform using LLMs, LangGraph, LangChain, RAG, and Model Context Protocol (MCP) to orchestrate multi-step agent workflows and context-aware reasoning.
- Designed and deployed backend services, retrieval pipelines, and agent-based execution frameworks using Docker, enabling seamless integration with external tools, knowledge bases, and enterprise workflows.

LLM Knowledge Base Platform

- Built an LLM-powered knowledge management platform with backend services, frontend applications, CLI tooling, and Obsidian integration for intelligent knowledge organization and retrieval.
- Implemented RAG pipelines, semantic search, and vector-based retrieval mechanisms to enable context-aware question answering across large collections of structured and unstructured documents.

Achievements

- Secured **2nd Place** at the **PGRS-RTGS Hackathon** organized by the Government of Andhra Pradesh
- Head of VR Club, VIT Chennai - led technical workshops and student projects in immersive technologies
- Awarded NCC Air Force Wing '**A**' Certificate from the Government of India, demonstrating excellence in cadet training and leadership
- Led a team on a Himalayan trek, managing navigation, logistics, and operations in high-altitude environments

Skills

Programming Languages: Python, C, C++, Java, R, SQL, JavaScript, C#, Bash

Machine Learning & AI: Deep Learning, Computer Vision, LangGraph, Retrieval-Augmented Generation, Agentic AI, Model Context Protocol

Databases & Data Engineering: MySQL, PostgreSQL, Database Design, SQL Optimization, ETL Pipelines, Apache Kafka

Data Analytics & Visualization: Pandas, NumPy, Power BI, Data Cleaning, Data Modeling

Backend & Full-Stack: Django, Flask, REST APIs, HTML, CSS, Bootstrap, React.js

Systems & DevOps: Docker, Linux, Git, GitHub

Cloud Platforms: AWS (EC2, S3), Cloud Deployment

Certifications & Professional Training

Machine Learning Specialization - DeepLearning.AI (Stanford) | Advanced Python Programming-University of Michigan | Software Engineering Virtual Programs - Walmart USA, J.P. Morgan, Cognizant | Computer Architecture - Saylor Academy | C/C++ & Java Programming (Intermediate)

Extracurricular Activities

NSS Volunteer (3 years) - community service & leadership initiatives | Trekker & Certified Mountaineer - ABVIMAS (Govt. of India)
| Technical workshops in AI tools, Linux & web development | Sports: Swimming, Skating, Football